

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Summer Examination – 2023****Course: B. Tech. Branch : Electronics & Computer Engineering Semester :IV****Subject Code & Name: Database Management System (BTECP402)****Max Marks: 60****Date: 15.07.2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level / CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) With help of a diagram, explain the difference between 2-tier and 3-tier architecture. Justify which architecture is better suited for web applications?	02/CO1	6
B) Describe with a diagram the level of abstractions available in database systems.	02/CO1	6
C) Describe the components of a a) Query Processor b) Storage Processor	02/CO1	6
Q. 2 Solve Any Two of the following.		12
i) Write SQL commands, i. To Create table for following schema: Student {Roll-No (PK), Name, Class, DOB} ii. To add new column percentage to student table. iii. To drop column DOB from student table.	03/CO2	6
ii) Consider an <i>EMPLOYEE</i> table having <i>Eid, Name, Age, Address, and Salary</i> . Using subqueries, write a query to i) Display all details where <i>Salary > 45000</i> ii) Delete all the records of employee whose <i>age < 20</i> . iii) Display name & salary of employee having second highest salary.	03/CO2	6
Explain Inner, Left Outer and Right Outer Joins with suitable example.	02/CO2	6

Q. 3 Solve Any Two of the following.

- A) What is Normalization? Explain various anomalies with suitable example. 02/CO3
- B) A company has following entities and respective attributes, where EmpPhone is a multivalued and Address is a composite attribute. Draw the ER-Diagram for the same;

Entity	Attributes
Employee	EmpId, EmpName, Gender, DoJ, Designation, EmpSal
Contact Details	EmpPhone, EmpMailID, Address
Department	DeptId, DepName, DepLocation, DepHead.
Project	ProjId, ProjName, ProjLocation, ProjDuration

03/CO3

C) Solve the following

- a. Find the Primary & Non-Primary Attributes of
 $R(A, B, C, D, E)$ FD: $\{A \rightarrow BCDE, BC \rightarrow ACE, D \rightarrow E\}$ 03/CO3
- b. Identify whether the given equations is in 2NF or not
 $R(A, B, C, D)$ FD: $\{AB \rightarrow CD, C \rightarrow A, D \rightarrow B\}$

Q.4 Solve Any Two of the following.

- A) Draw a state diagram and discuss the typical states that a transaction goes through during execution. 03/CO4
- B) What is concurrency control? Explain lock based protocols for concurrency control. 02/CO4
- C) Describe shortly 02/CO4
- a) Database Recovery Techniques
 - b) Causes of Database failures

Q. 5 Solve Any Two of the following.

- A) State the CAP theorem. Explain each term of CAP with suitable example. 01/CO5
- B) Describe the concept of Document and Collection in MongoDB. 02/CO5
- C) For MongoDB, write a query to 03/CO5
- a) Create a collection named *Library*
 - b) Create a document named using data given below
BN101, "Database System Concepts", "Henry Korth", 800.50.
 - c) Insert the following book details using single query.
BN102, "Core Java", "Nageswara Rao", 650.75.
BN103, "Let Us C", "Balaguruswamy", 400.50.
BN104, "DBMS", "G.D. Patil", 250.5
 - d) Retrieve / Display all the above documents
 - e) Update the Author name to "Y Kanetkar" for BookId BN103
 - f) Delete a document with BookId BN104.

*** End ***